

5.25 Solution: (a) Using the configuration shown in the scanned solution, we have

$$r_1^2 = d^2 - bd \cos \alpha + \frac{b^2}{4}, \quad r_2^2 = d^2 + bd \cos \alpha + \frac{b^2}{4}.$$

For a current at the origin in the $-\hat{z}$ -direction, the vector potential at a distance r away from it is

$$\mathbf{A} = \hat{z} \frac{\mu_0 I_2}{2\pi} \log r.$$

Since the current I_1 in the side with length a that is closest to the wire at origin has the same direction, while the other side with length a is opposite, the interaction energy can be easily calculated as

$$\begin{aligned} W_{12} &= -\frac{\mu_0 I_1 I_2 a}{2\pi} \log r_1 + \frac{\mu_0 I_1 I_2 a}{2\pi} \log r_2 = \frac{\mu_0 I_1 I_2 a}{2\pi} \log \left(\frac{r_2}{r_1} \right) \\ &= \frac{\mu_0 I_1 I_2 a}{4\pi} \log \left(\frac{4d^2 + b^2 + 4bd \cos \alpha}{4d^2 + b^2 + 4bd \cos \alpha} \right). \end{aligned}$$

(b) We want to use Eq. (5.151) to calculate the force, but the energy at this moment only depends on vertical distance from the current at the origin to the rectangular loop. To this end, let us shift the loop an ϵ distance away from the plane containing the wire and the center of the loop. In this case,

$$r_1^2 = \left(d - \frac{b}{2} \cos \alpha \right)^2 + \left(\epsilon + \frac{b}{2} \sin \alpha \right)^2, \quad r_2^2 = \left(d + \frac{b}{2} \cos \alpha \right)^2 + \left(\epsilon - \frac{b}{2} \sin \alpha \right)^2,$$

and the interaction energy, as a function of d and ϵ is

$$W_{12} = \frac{\mu_0 I_1 I_2 a}{4\pi} \log \left(\frac{(2d + b \cos \alpha)^2 + (2\epsilon - b \sin \alpha)^2}{(2d - b \cos \alpha)^2 + (2\epsilon + b \sin \alpha)^2} \right).$$

Now, the force can be calculated as

$$\begin{aligned} \mathbf{F} &= \left(\frac{\partial W_{12}}{\partial \epsilon}, \frac{\partial W_{12}}{\partial d} \right) \Big|_{\epsilon=0} \\ &= -\frac{\mu_0 I_1 I_2 a}{2\pi} \frac{1}{(d^2 + b^2/4)^2 - b^2 d^2 \cos^2 \alpha} \begin{pmatrix} (d^2 + b^2/4)b \sin \alpha \\ (d^2 - b^2/4)b \cos \alpha \end{pmatrix}^\top. \end{aligned}$$

The force can also be explicitly calculated for this configuration. See scanned solution for detail.

(c) Let us first set up the configuration from scratch. Consider a ring originally in the xz -plane, with center at the origin. Points on this ring can be parameterized by angle θ . Then, for a point on the ring, we have its location and the corresponding current density at this point as

$$\mathbf{r} = (a \cos \theta, 0, a \sin \theta), \quad \mathbf{I}_1 = -I_1 \hat{\theta} = I_1 (\sin \theta, 0, -\cos \theta).$$

Now, rotate this ring around the z -axis by an angle of $\pi/2 - \alpha$, then

$$\mathbf{r} = (a \cos \theta \sin \alpha, -a \cos \theta \cos \alpha, a \sin \theta), \quad \mathbf{I}_1 = I_1 (\sin \theta \cos \alpha, -\sin \theta \sin \alpha, -\cos \theta).$$

Finally, move the center of the ring along y -axis by a distance of d , and

$$\mathbf{r} = (a \cos \theta \sin \alpha, d - a \cos \theta \cos \alpha, a \sin \theta),$$

while $\mathbf{I}_1$ stays the same. In fact, our experience in part (b) suggests that we should also move the ring by an ϵ of distance in x -direction. Therefore, points on the ring will be

$$\mathbf{r} = (\epsilon + a \cos \theta \sin \alpha, d - a \cos \theta \cos \alpha, a \sin \theta),$$

and we will set ϵ to 0 in the final step.

From part (a), the interaction energy is

$$W_{12} = \oint \mathbf{J}_1 \cdot \mathbf{A} dl = \oint (-\hat{z} I_1 \cos \theta) \cdot \left(\hat{z} \frac{\mu_0 I_2}{2\pi} \log r \right) dl = -\frac{\mu_0 I_1 I_2 a}{2\pi} \int_0^{2\pi} \cos \theta \log r d\theta,$$

where r is the distance from a point on the ring to the z -axis,

$$r^2 = (\epsilon + a \cos \theta \sin \alpha)^2 + (d - a \cos \theta \cos \alpha)^2 = |d + i\epsilon - a \cos \theta e^{-i\alpha}|^2,$$

and

$$\log r = \log |d + i\epsilon - a \cos \theta e^{-i\alpha}| = \text{Re} [\log (d + i\epsilon - a \cos \theta e^{-i\alpha})].$$

This is the key to the final desired form of solution to this problem.

Now,

$$W_{12} = -\frac{\mu_0 I_1 I_2 a}{2\pi} \text{Re} \left[\int_0^{2\pi} \cos \theta \log (d + i\epsilon - a \cos \theta e^{-i\alpha}) d\theta \right],$$

and we need to perform the integral,

$$\begin{aligned} Q &= \int_0^{2\pi} \cos \theta \log (p - q \cos \theta) d\theta \\ &= \sin \theta \log (p - q \cos \theta) \Big|_0^{2\pi} - \int_0^{2\pi} \frac{q \sin^2 \theta}{p - q \cos \theta} d\theta \\ &= -\frac{2\pi p}{q} \left(1 - \sqrt{1 - \frac{q^2}{p^2}} \right), \end{aligned}$$

where we have used Gradshteyn and Ryzhik, 8th ed., 3.644.1, pg. 404 (not found), with $|q| < |p|$.

(Interestingly, I asked ChatGPT to calculate the integral for me, but it told me that the integrand is an *odd* function, so the result is 0. But, if I use the integral after integration by parts, it can give me the correct result.)

Therefore, with nonzero ϵ , $p = d + i\epsilon$ and $q = ae^{-i\alpha}$,

$$W_{12} = \mu_0 I_1 I_2 \cdot \text{Re} \left\{ (d + i\epsilon) \left[e^{i\alpha} - \sqrt{e^{2i\alpha} - \frac{a^2}{(d + i\epsilon)^2}} \right] \right\}.$$

Setting ϵ to 0, we will obtain the desired result.

The force can be calculated in the same way as in part (b),

$$\begin{aligned} \mathbf{F} &= \left(\frac{\partial W_{12}}{\partial \epsilon}, \frac{\partial W_{12}}{\partial d} \right) \Big|_{\epsilon=0} \\ &= \mu_0 I_1 I_2 \begin{pmatrix} -\text{Im} \left[e^{i\alpha} - \frac{e^{2i\alpha}}{\sqrt{e^{2i\alpha} - a^2/d^2}} \right] \\ \text{Re} \left[e^{i\alpha} - \frac{e^{2i\alpha}}{\sqrt{e^{2i\alpha} - a^2/d^2}} \right] \end{pmatrix}^T. \end{aligned}$$

(d) In the limit $d \gg a, b$, we have

$$\sqrt{e^{2i\alpha} - \frac{a^2}{d^2}} \approx e^{i\alpha} - \frac{a^2}{2d^2} e^{-i\alpha},$$

and

$$\log \left(d^2 \pm bd \cos \alpha + \frac{b^2}{4} \right) \approx \log d^2 \pm \frac{b}{d} \cos \alpha.$$

Then, for the circular loop,

$$W_{12} = \mu_0 I_1 I_2 d \cdot \operatorname{Re} \left[\frac{a^2}{2d^2} e^{-i\alpha} \right] = \frac{\mu_0 I_2}{2\pi d} \cdot \pi a^2 I_1 \cdot \cos \alpha = \mathbf{m} \cdot \mathbf{B},$$

and for the rectangular loop,

$$W_{12} = \frac{\mu_0 I_1 I_2 a}{4\pi} \cdot \frac{2b}{d} \cos \alpha = \frac{\mu_0 I_2}{2\pi d} \cdot ab I_1 \cdot \cos \alpha = \mathbf{m} \cdot \mathbf{B}.$$